

Causal Inference with Ordinal Outcomes: Flexible Estimation Using Normalizing Flows

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Motivation and Target Estimands

Ordinal responses encode rankings, but not distances between categories. Assigning scores and estimating a mean difference therefore adds an untestable measurement assumption.

Target causal effects on the observed response distribution,
not coefficients on an arbitrary latent scale.

For response category $j = 1, \dots, J$, define

$$\pi_j(d) = \Pr\{Y_i(d) = j\}, \quad \Delta_j = \pi_j(1) - \pi_j(0).$$

Cumulative effects are

$$\Delta_{\geq j} = \Pr\{Y_i(1) \geq j\} - \Pr\{Y_i(0) \geq j\}.$$

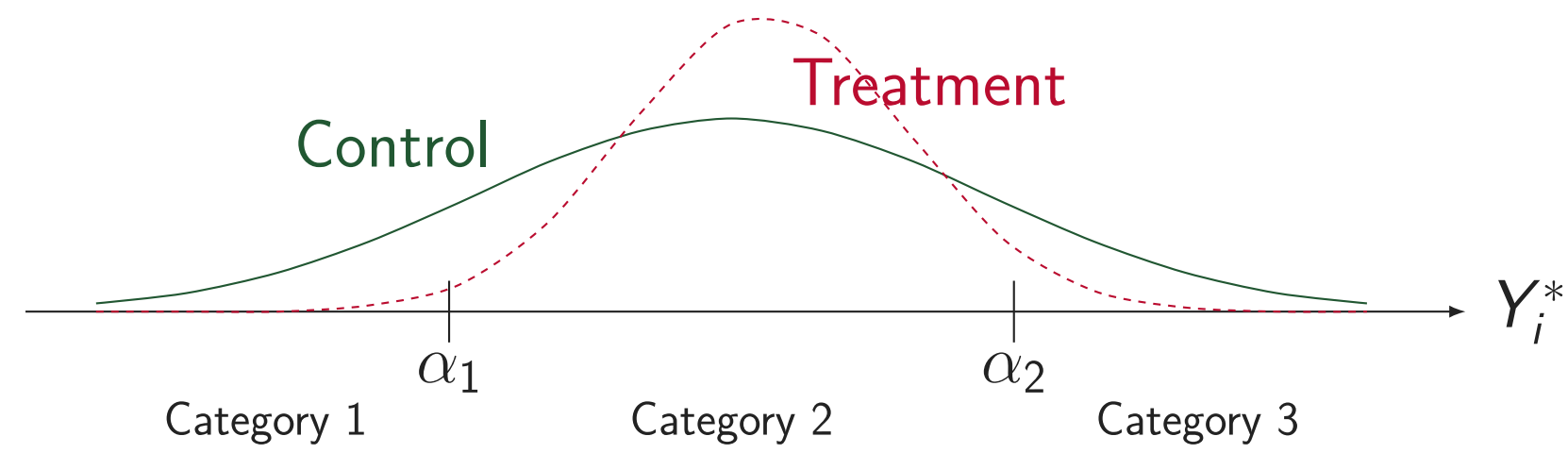
These quantities are invariant to admissible transformations of the latent scale.

Single-Index Representation

Observed categories arise from a threshold-crossing model with latent outcome Y_i^* :

$$Y_i = j \iff \alpha_{j-1} < Y_i^* \leq \alpha_j, \quad Y_i^* = m(D_i, X_i) + \varepsilon_i.$$

A structured specification uses $m(D_i, X_i) = \tau D_i + h(X_i)$. The coefficient τ shifts a latent index, but its magnitude depends on location and scale normalizations. It is not itself an observed-scale causal effect.



Why Cardinalization Does Not Identify the Effect

Let $c(1) < \dots < c(J)$ be any numerical coding of the categories. The corresponding mean effect is

$$\tau_c = \mathbb{E}[c(Y_i(1)) - c(Y_i(0))] = \sum_{j=1}^J c(j) \Delta_j.$$

Because ordinal data do not identify the spacing in $c(\cdot)$, neither the magnitude nor, in general, the sign of τ_c is ordinally identified.

Cardinalization result

Different monotone codings can imply different causal conclusions while preserving exactly the same observed ordering.

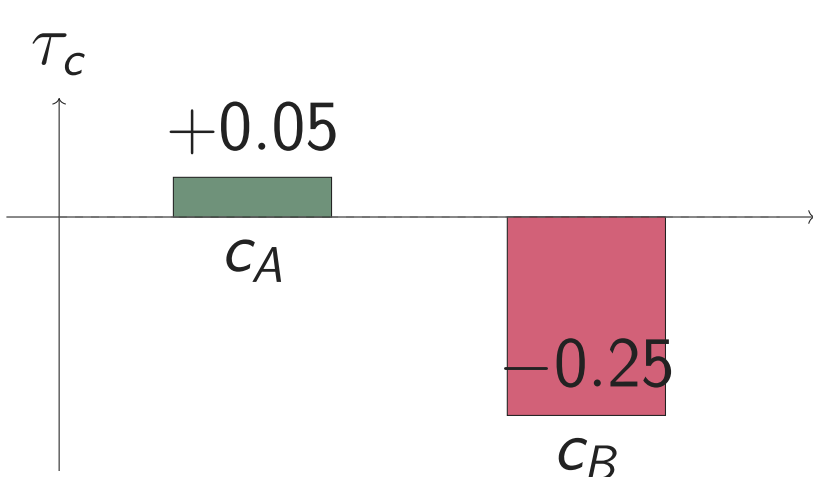
Toy Example: The Sign Flips

Suppose treatment redistributes probability across three ordered categories:

$$(\Delta_1, \Delta_2, \Delta_3) = (-0.15, 0.25, -0.10).$$

Then two admissible codings yield opposite signs:

$$\begin{aligned} c_A = (1, 2, 3) : \tau_{c_A} &= 1 \cdot (-0.15) + 2 \cdot 0.25 + 3 \cdot (-0.10) = +0.05, \\ c_B = (1, 2, 6) : \tau_{c_B} &= 1 \cdot (-0.15) + 2 \cdot 0.25 + 6 \cdot (-0.10) = -0.25. \end{aligned}$$



Identification of Distributional Effects

Under standard identification conditions, the full distribution of potential ordinal outcomes is identified.

Assumptions (SUTVA, Conditional Ignorability, and Positivity):

$$Y_i(d) \perp D_i \mid X_i \quad \text{and} \quad 0 < \Pr(D_i = d \mid X_i) < 1 \quad \forall d \in \{0, 1\}.$$

We identify the marginal potential category probabilities via standardization:

$$\pi_j(d) = \mathbb{E}_X[\Pr(Y_i = j \mid D_i = d, X_i)].$$

- For randomized experiments, random assignment supplies ignorability. Thus the causal target is identified.
- the remaining challenge is estimating the conditional ordinal response distribution.

Estimation Problem

For a latent index $m(d, x)$ and conditional error CDF $F_{\varepsilon|D,X}$,

$$\begin{aligned} p_j(d, x) &= F_{\varepsilon|D,X}(\alpha_j - m(d, x) \mid d, x) \\ &\quad - F_{\varepsilon|D,X}(\alpha_{j-1} - m(d, x) \mid d, x). \end{aligned}$$

- If the normal/logistic error family and index specification are correct, ordered probit/logit are efficient and adequate.
- However, we cannot *ex ante* sure whether they are correct.

Two Threats to Estimation

1. Error distribution

Normal or logistic errors cannot represent skewness, multimodality, polarization, or treatment-dependent dispersion.

2. Single-index structure

$Y_i^* = \tau D_i + h(X_i) + \varepsilon_i$ may fail when treatment changes shape rather than only location.

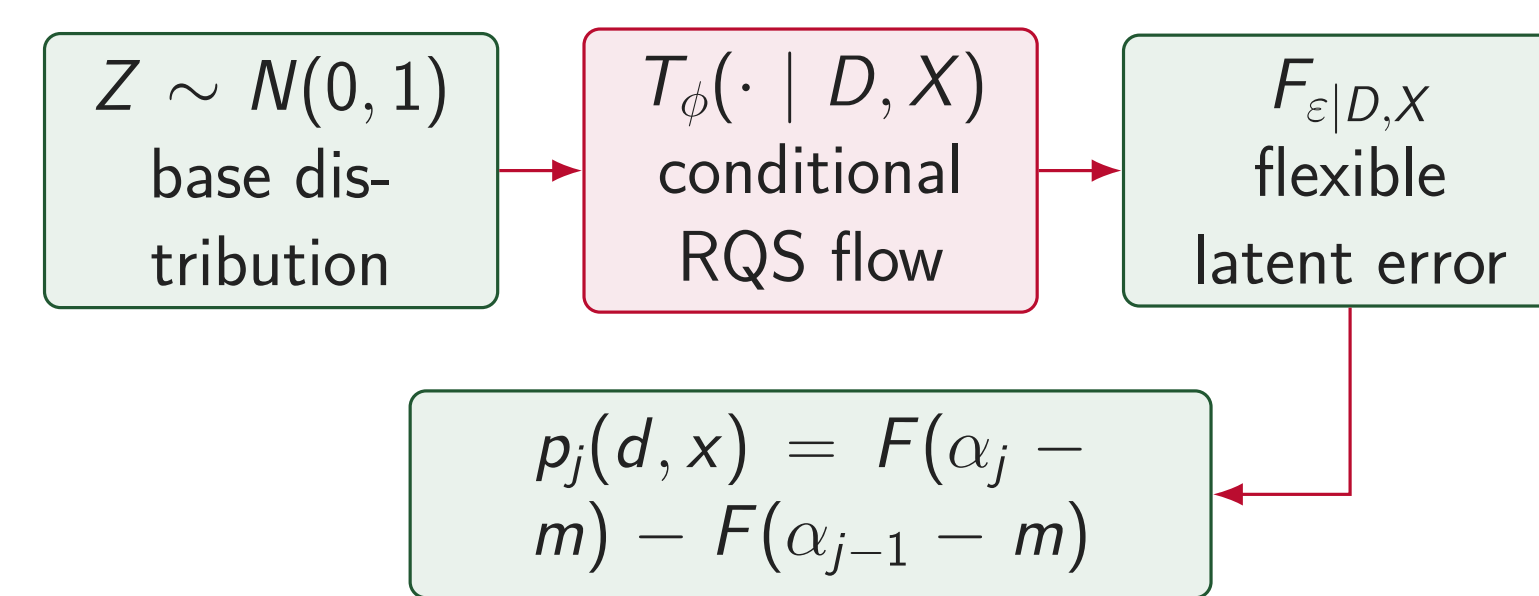
Conditional Normalizing Flows

Let $Z_i \sim N(0, 1)$ and transform it using an invertible conditional map:

$$\varepsilon_i = T_\phi(Z_i \mid D_i, X_i), \quad Z_i = T_\phi^{-1}(\varepsilon_i \mid D_i, X_i).$$

The implied CDF is available as:

$$F_{\varepsilon|D,X}(e \mid d, x) = \Phi(T_\phi^{-1}(e \mid d, x)).$$



- Conditional Normalizing Flow models guarantees more flexible likelihood-based distribution estimation.
- Monotone rational-quadratic splines (Durkan et al., 2019) retain invertibility, analytical CDF evaluation.

Structured and Model-Free Specifications

	Structured flow	Model-free flow
Systematic component	$\tau d + h_\eta(x)$	learned nonparametrically
Latent distribution	conditional flow	conditional flow
Relaxes	error family	error family and index
Strength	efficiency	robustness
Cost	residual index risk	greater data demand

Both specifications target the same standardized category-specific and cumulative effects.

Likelihood and Consistency

The parameters $\Theta = \{\tau, h, \phi, \alpha\}$ maximize the sample log-likelihood:

$$\ell_n(\Theta) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^J \mathbf{1}(Y_i = j) \log p_j(D_i, X_i; \Theta).$$

Assumption C.3 (Flow Approximation & Identification): The flow family $\{F_\phi\}$ is rich enough to approximate the true error CDF F_0 arbitrarily well. The parameters are identified up to a positive affine transformation $(a + c\alpha, c\beta, c\tau)$ with $c > 0$.

Assumption C.4 (Regularity Conditions): The parameter space is compact, the likelihood satisfies standard continuity and differentiability conditions, and the model is correctly specified.

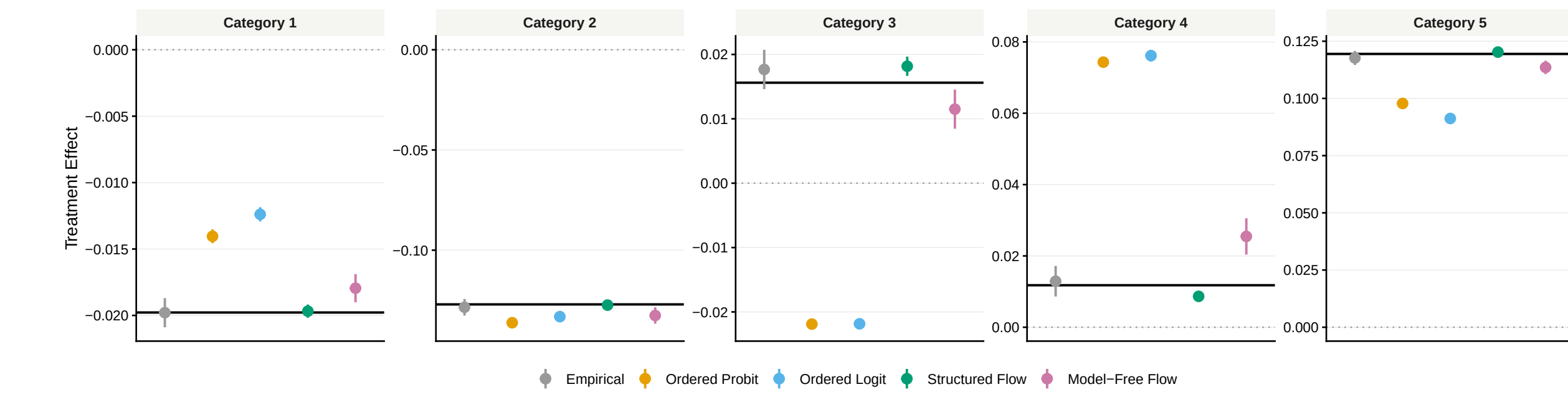
Theorem C.2 (Consistency of the Flow Estimator)

Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions C.3–C.4 hold. Then, as $n \rightarrow \infty$, the maximum likelihood estimator is consistent up to a positive affine transformation:

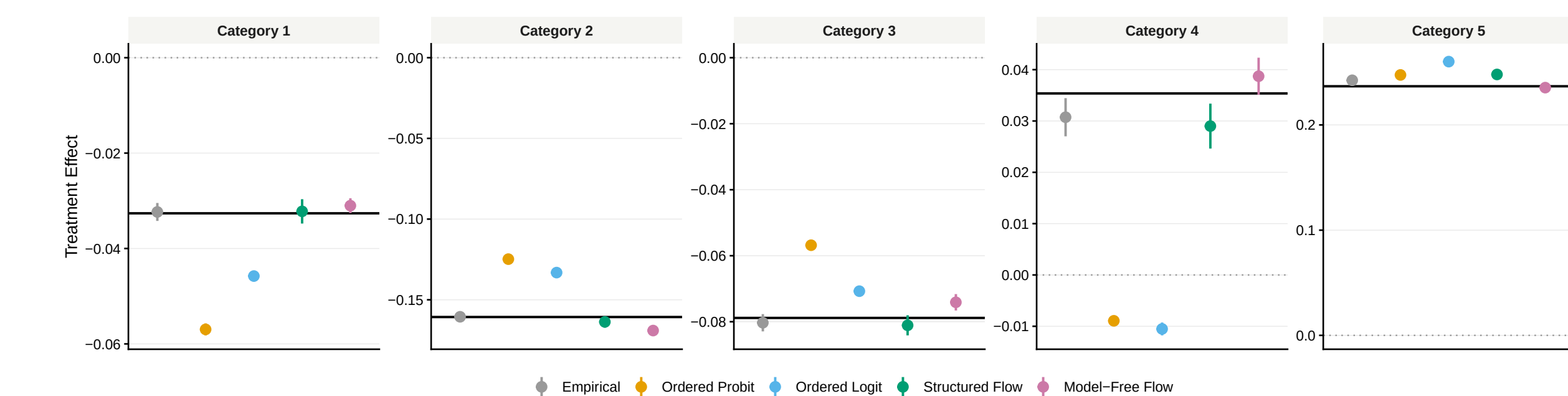
$$\|\hat{\Theta} - \Theta_0\| = o_p(1).$$

Monte Carlo Simulation

- **Polarization:** a bimodal latent error distribution.
- **Index failure:** nonlinear treatment-response structure.



Category-specific effects under a polarized latent distribution ($N = 2000$).



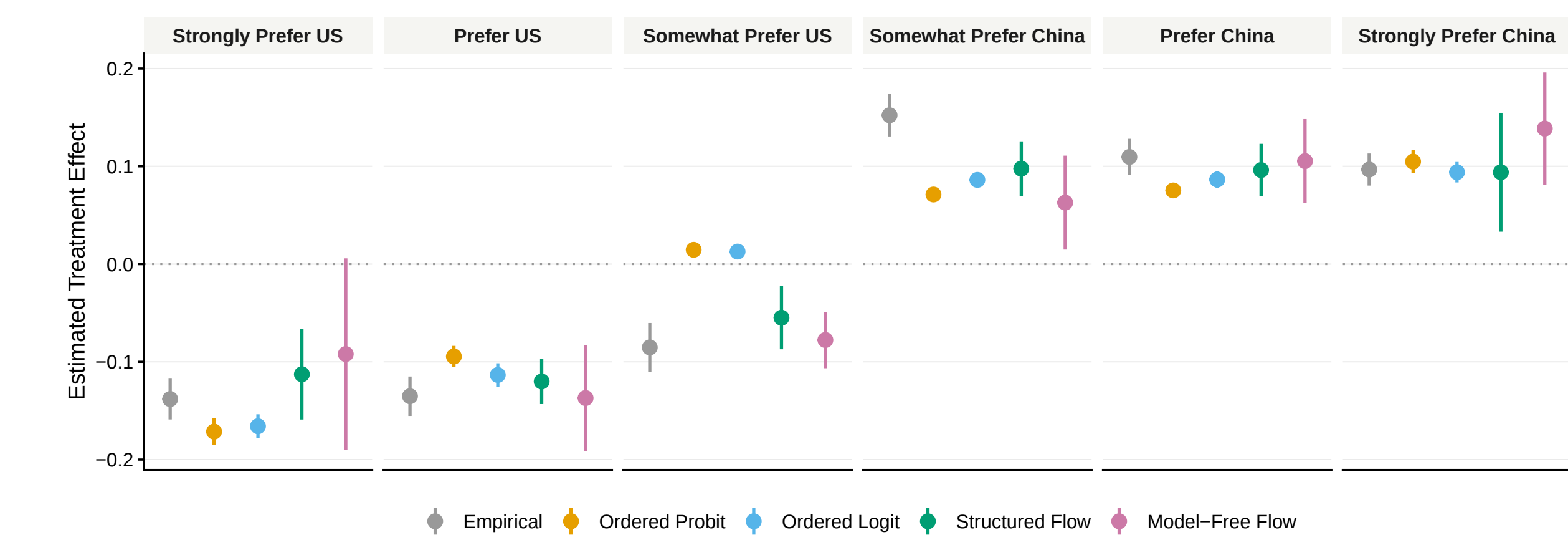
Category-specific effects under a nonlinear latent distribution ($N = 2000$).

- Parametric error misspecification distorts category-level effects.
- Structured flows improve robustness with little efficiency loss.
- Model-free flows are most flexible but noisier in finite samples.

Application: Authoritarian Propaganda

Replicating Mattingly et al.(2025)'s survey experiment on how Chinese state media affects public opinion on relative preference for the United States versus China.

- **Parametric Models (Probit/Logit)** Overestimating the conversion of citizens into extreme supporters (*Strongly Prefer China*).
- **Flexible Flow Estimators** show the propaganda's primary impact is concentrated in **intermediate categories**.
- **Political Science Implication:** Authoritarian propaganda is persuasive not by creating extreme ideologues, but by softening opposition and winning over moderate/indifferent citizens.



Conclusion

1. **Cardinalized mean effects are not ordinally identified.**
2. **Category-specific and cumulative probability effects are causally interpretable under standard identification assumptions.**
3. **Conditional flows separate substantive conclusions from restrictive error-distribution and index assumptions.**

Report distributional effects and compare structured and model-free estimators when ordered logit/probit assumptions are doubtful.

Contact and Materials

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Code & Data: https://github.com/chanhyuk58/ordinal_outcome